APPENDIX A

Logic and Proof Methods

A.1 Logic

This appendix introduces some foundations of mathematical logic. We will study propositions, quantifiers, and logical connectives, and see how they are used to construct proofs and assess mathematical claims.

A.1.1 Propositions

A proposition is a statement to which exactly one of the truth values ``true'' and ``false'' can be assigned unambiguously. Examples of propositions include:

- Rome is the capital of Italy. (A true proposition.)
- $3^2 = 8$. (A false proposition.)

Consider the statement that the number of grains of sand on Earth exceeds 10^{24}. We may not know its truth value, and checking it would be difficult. Nevertheless, under an agreed meaning of the statement, it is either true or false; our lack of knowledge does not itself prevent it from being a proposition.

Examples of expressions that are not propositions include:

- How are you? (A question is not a proposition.)
- Please close the door! (An instruction is not a proposition.)
- $x^3 > x$. (No truth value is determined until the meaning or quantification of x has been specified.)

The sentence ``The Mona Lisa is a beautiful painting'' expresses a subjective judgement. To treat it as a formal proposition, one would first need to specify a precise meaning of ``beautiful.'' In mathematical logic, we work with statements whose meanings and domains of discourse have been specified.

A.1.2 Quantifiers

- Consider ``He is the prime minister.'' Without specifying the referent, this does not determine a proposition. Similarly, $n > 4$ does not determine a truth value when n is unspecified. To bind variables in statements, we use two logical symbols called **quantifiers**.

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RECALL AND TOOLS

- **For every**, denoted by $\forall$. For example, ``Every natural number is greater than 4'' is written as
$$\forall n \in \mathbb{N}\ (n > 4).$$
- **There exists**, denoted by $\exists$. For example, ``There exists a natural number greater than 4'' is written as
$$\exists n \in \mathbb{N}\ (n > 4).$$

Specify the domain of each quantifier, either explicitly or through a clearly established context. The following example explains why this matters.

- The expression $\forall x\ x^3 \geq x$ is incomplete without an agreed domain. The operations and comparison may not even be defined for the objects under consideration. In contrast, the following are propositions because the domain is stated:
  - $\forall x \in \mathbb{N}\ x^3 \geq x$ is true, since $x^3 \geq x$ for every natural number x.
  - $\forall x \in \mathbb{Q}\ x^3 \geq x$ is false. Choosing $x = \frac{1}{3}$ gives $x^3 = \frac{1}{27} < \frac{1}{3} = x$, a counterexample.

EXERCISE **A.1**

Please write the following statement in mathematical notation: a set is contained in another set if and only if every element of the first set also belongs to the second set.

WORKED SOLUTION

$$A \subseteq B \iff \forall a \in A,\ a \in B.$$

We've reached the finish. Victory!

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Logical Connectives: Conjunction, Disjunction, and Negation

Conjunction ($\wedge$): both conditions hold.

- Saying that ``our room is large and well lit'' means that ``our room is large'' *and* ``our room is well lit.'' The conjunction is true exactly when both statements are true. If either statement is false, the conjunction is false.

Truth table: The table gives the truth value of $A \wedge B$ for every possible pair of truth values of A, B.

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction ($\vee$): logical ``or.''

- $A \vee B$ is true when at least one of A, B is true, and false when both are false. This is **inclusive** ``or'': it permits both statements to be true.

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Truth table:

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

Negation ($\neg$): reversing a truth value.

- For example, the negation of ``Every multiple of $6$ is a multiple of $3$'' is ``Not every multiple of 6 is a multiple of 3,'' or equivalently, ``There exists a multiple of 6 that is not a multiple of 3.'' A proposition and its negation have opposite truth values: one is true exactly when the other is false.

Truth table:

A	$\neg A$
T	F
F	T

EXERCISE A.2

Please use logical quantifiers to write the following proposition and its negation: there exists a natural number M such that $n \leq M$ for every natural number n.

WORKED SOLUTION

The proposition is

$$\exists M \in \mathbb{N}\, \forall n \in \mathbb{N},\ n \leq M.$$

Its negation is

$$\forall M \in \mathbb{N}\, \exists n \in \mathbb{N},\ n > M.$$

Yes! Victory!

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Implication ($\Rightarrow$) and Equivalence ($\Leftrightarrow$)

**Implication ($\Rightarrow$): ``if … then ….''** Consider the statement ``If Tamar received a final grade of at least 75, then Tamar passed the course.'' Let A denote the assumption and B the conclusion.

- If A is true and B is true, the implication is true.
- If A is true and B is false, the implication is false.
- If A is false and B is true, the implication is true.
- If A is false and B is false, the implication is true.

This example illustrates the meaning of implication.

The implication is **false** only when A, the assumption, is true and B, the conclusion, is false. In every other case it is true.

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Truth table:

A	B	$A \Rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

Example: The proposition

$$4 = 7 \Rightarrow 6 = 9$$

is **true**: its assumption is false, even though its conclusion is also false.

Important remarks. For a statement of the form ``If $P$, then $Q$,'' we ask whether $Q$ must hold whenever $P$ holds. Everyday language often uses ``if … then …'' causally (for example, ``If you eat, you will feel full''). Formal implication has the truth-table meaning given above; it need not express a causal connection.

Negating an implication. The implication fails exactly when P is true and Q is false:

$$\neg(P \Rightarrow Q) \equiv P \wedge \neg Q.$$

In words: The statement $P \Rightarrow Q$ says that whenever P is true, Q is true as well. To negate it, identify the case in which it fails: P is true but Q is false.

Equivalence ($\Leftrightarrow$): ``if and only if.''

- $P \Leftrightarrow Q$ is true exactly when P, Q have the same truth value: either both are true or both are false.

Truth table:

A	B	$A \Leftrightarrow B$
T	T	T
T	F	F
F	T	F
F	F	T

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Proof Techniques

A.1.3 Proof Templates

NOTE

The following templates summarize proof methods that have already appeared and will be used throughout the course.

Universal Statements

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RECALL AND TOOLS

A universal statement asserts that every object satisfying a property p also satisfies a property q. It can be written as an implication: if an object satisfies p, then it satisfies q.
To prove the statement, take an arbitrary object and assume only that it satisfies p. Show that q follows. Be careful not to impose additional assumptions on the object.
To disprove the statement, find an object satisfying p but not q. One such counterexample is enough to disprove a universal statement.

EXERCISE A.3

Please prove that

$$\forall x \in \mathbb{R},\ (x > 3) \Rightarrow (x^2 > 9).$$

WORKED SOLUTION

Let $x \in \mathbb{R}$ satisfy $x > 3$. Since x is positive, multiply both sides of the inequality by x without reversing its direction:

$$x \cdot x > 3 \cdot x.$$

Hence

$$x^2 > 3x.$$

Also, $x > 3$ implies $3x > 9$. By transitivity,

$$x^2 > 9.$$

We have therefore shown that

$$(x > 3) \Rightarrow (x^2 > 9).$$

Because x was arbitrary, the conclusion holds universally:

$$\forall x \in \mathbb{R},\ (x > 3) \Rightarrow (x^2 > 9).$$

V-I-C-T-O-R-Y!

EXERCISE A.4

Please prove by contradiction: if $A \setminus B = \varnothing$, then $A \subseteq B$.

WORKED SOLUTION

Assume that $A \setminus B = \varnothing$ and, for a contradiction, that $A \not\subseteq B$. Then some $x \in A$ satisfies $x \notin B$, so $x \in A \setminus B$. This contradicts the assumption that the difference is empty. Therefore $A \subseteq B$.

Victory!

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EXERCISE A.5

Please disprove the following statement by a counterexample: $x^3 > x$ for every $x \in \mathbb{R}$.

WORKED SOLUTION

The statement is false. Take $x = \frac{2}{3}$. Then

$$\left(\frac{2}{3}\right)^3 = \frac{8}{27} < \frac{2}{3},$$

so the proposed strict inequality fails.

And we're done. Victory!

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